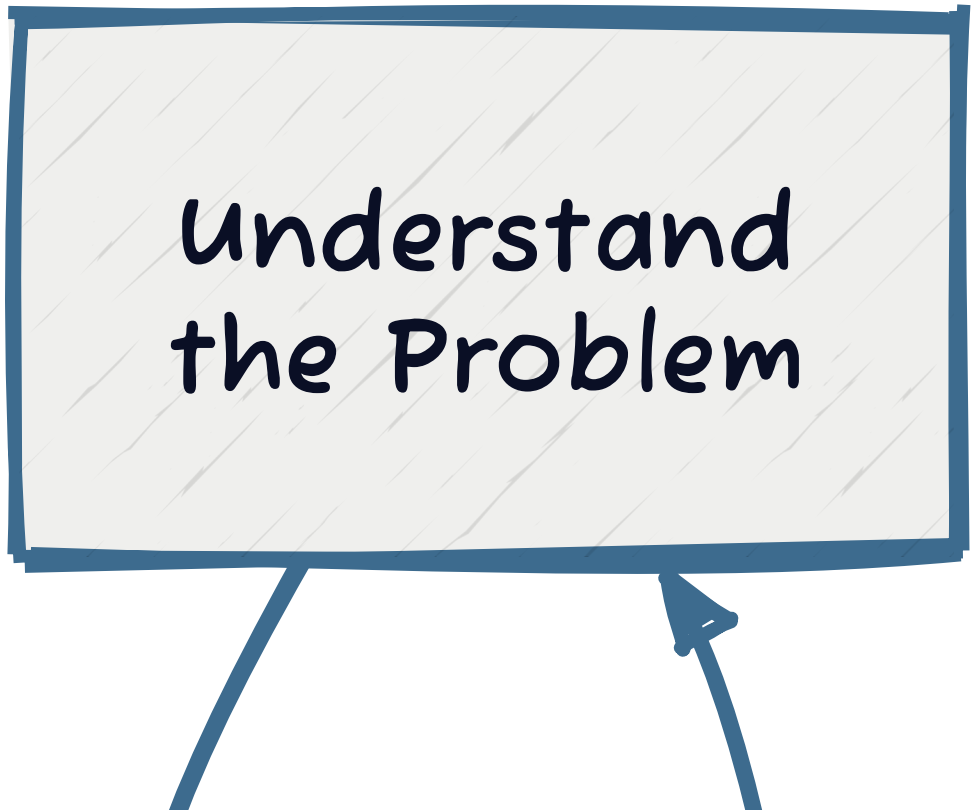
Day 04

The Modeling Project

MSU REU Machine Learning Short Course

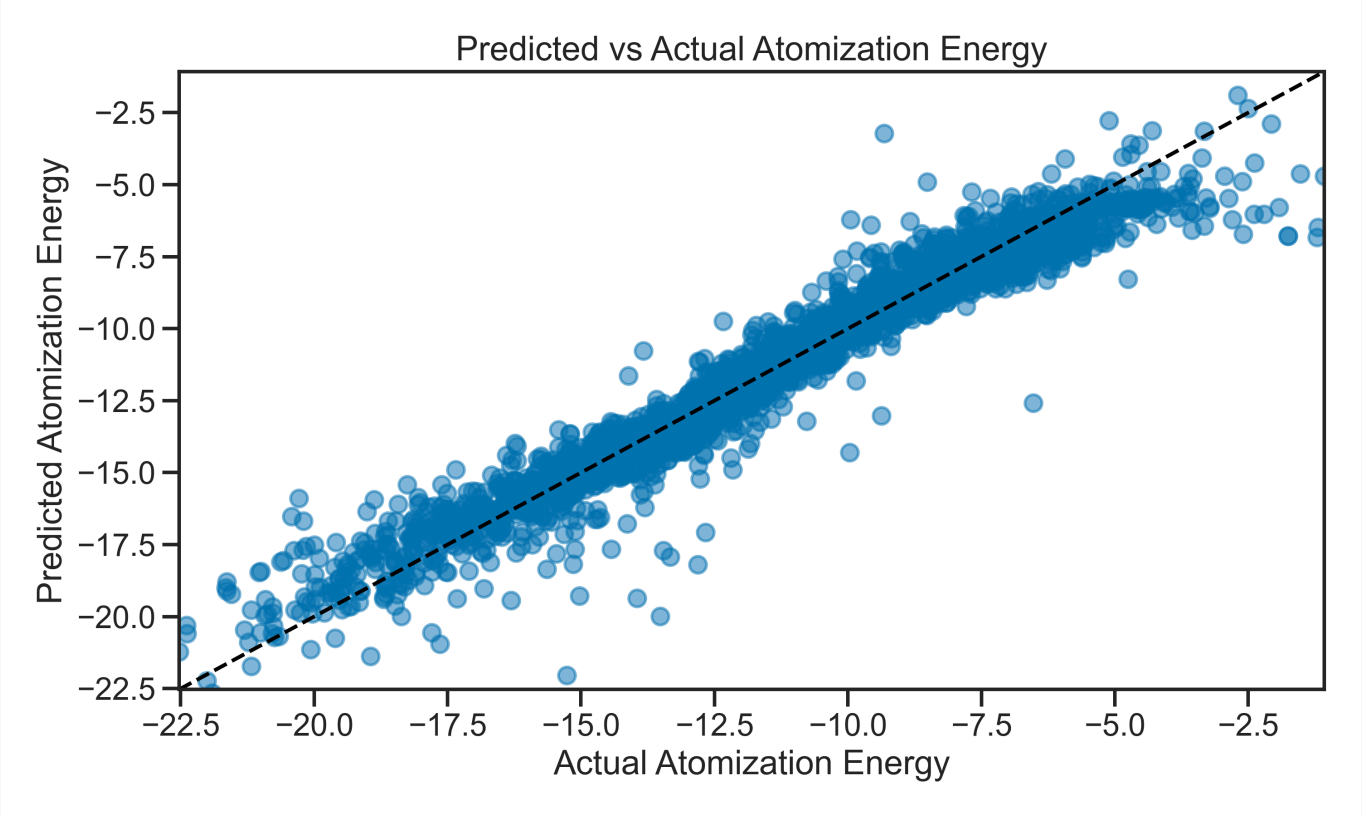
The Full ML Loop

Days 1–3: Each piece separately. **Today: You run the entire loop yourself.**



Today's Dataset — Molecular Atomization Energy

Predict the energy required to pull a molecule apart, from its atomic structure.



1275 features from the Coulomb matrix representation of each molecule.

Real-world challenge: Most features are noise. Your job: find the signal.

Why This Dataset is Different

	SDSS (Days 1–3)	Molecular Energy (Today)
Size	100k objects	1275 features each
Features	17 (clean, well-understood)	1275 (mostly noise)
Problem	Which features matter for classification?	How do you even start?

Day 1–3: You could visualize all features, understand relationships.

Today: 1275-dimensional space. You can't visualize it. You need systematic methods.

The Curse of Dimensionality

What happens when you have many features?

- **More features = more parameters** — easier to fit noise, not just signal
- **Overfitting risk increases** — with 1275 features, a linear model can fit random data perfectly
- **Computational cost** — training takes longer
- **Interpretability collapse** — what do 1275 learned weights mean?

The goal: Remove noise, keep signal. Go from 1275 → ~50 features that actually matter.

Overfitting vs Underfitting: The Goldilocks Problem

Underfitting: Model too simple. Misses real patterns. High bias.

Overfitting: Model memorizes noise. Fails on new data. High variance.

Goldilocks zone: Model is complex enough to capture signal, but simple enough to generalize.

Why Start with a Terrible Model?

Build the simplest possible model first. Expect it to fail.

This is your **diagnostic baseline**. Failure teaches you where to improve.

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LinearRegression()
model.fit(X_train, y_train)
r2 = r2_score(y_test, model.predict(X_test))
print(f"R² = {r2:.3f}")  # probably terrible (e.g., 0.15)
```

Feature Selection: Art and Science

Not all 1275 features carry signal. Systematic methods find which ones to keep.

Approaches:

- **Recursive Feature Elimination (RFE):** Iteratively remove least important features
- **Domain knowledge:** Use physics intuition (some features are obviously useless)
- **Correlation threshold:** Remove redundant features (e.g., if two features are 99% correlated, keep one)

> See: Recursive Feature Elimination

Cross-Validation: More Robust Confidence

One train/test split = one number = could be lucky or unlucky.

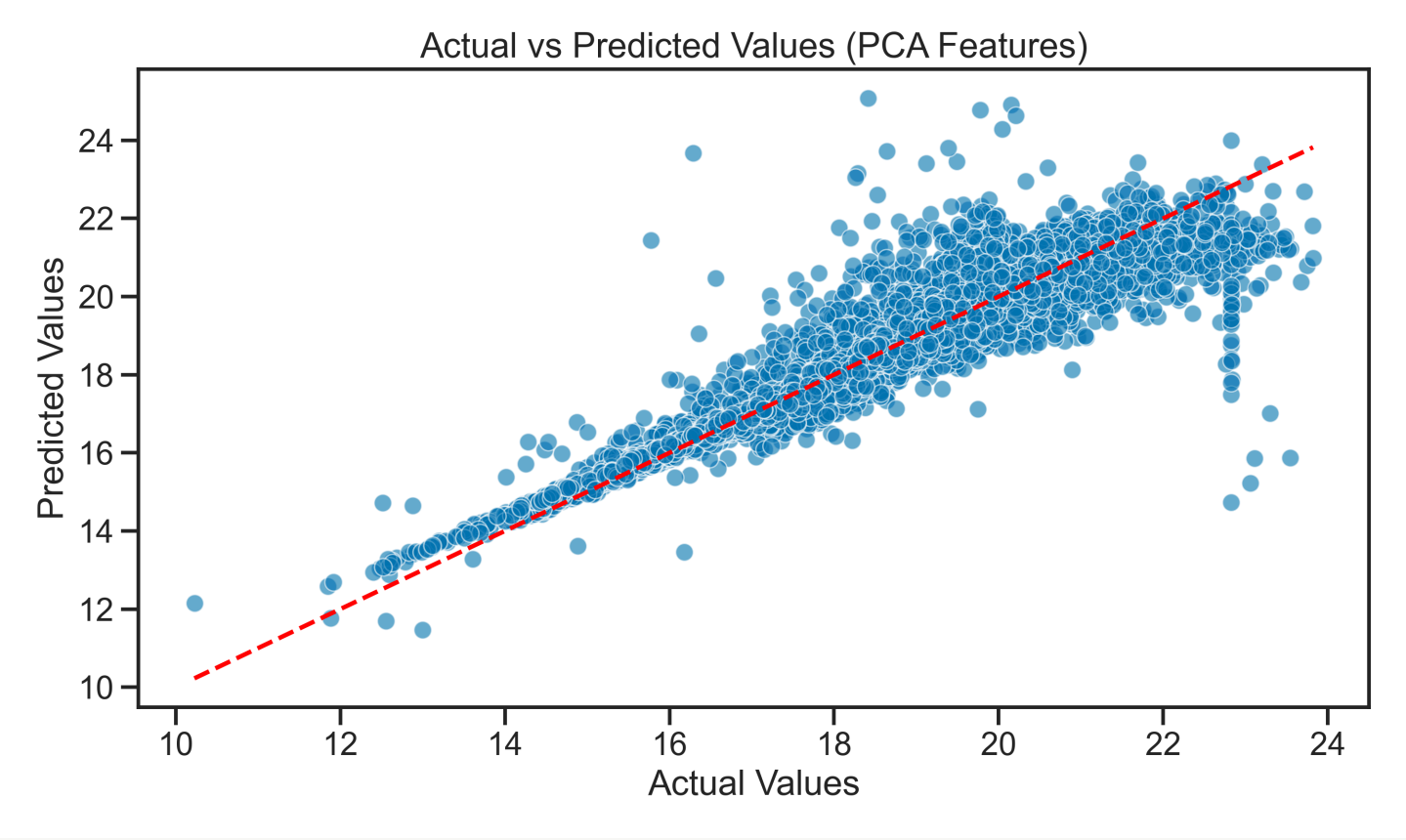
K-fold cross-validation: Run the model k times, get k scores, average them.

Example with k=5:

- Fold 1: Train on 80%, test on 20%
- Fold 2: Train on different 80%, test on different 20%
- ... (repeats 5 times)
- Average the 5 R^2 scores

Benefit: More robust. One unlucky split won't fool you. You get a distribution, not a single number.


```
from sklearn.model_selection import cross_val_score
```



What's happening? PCA finds new axes (components) where data spreads out most. First 50 PCs often capture 95% of variance.

Trade-off:

- Lose interpretability (what does "PC17" mean?)

The Iteration Loop: How Real Data Science Works

This is not a linear process. You build, evaluate, diagnose, fix, repeat.

Each iteration teaches you:

- Where your model fails (residuals)
- Which features matter (coefficients, feature importance)
- How confident you should be (cross-validation scores)

When to stop? When you understand your model and its limitations, even if it's not perfect.

Common Pitfalls: What Will Bite You

1. **Using all features** — Linear Regression with 1275 features will overfit
2. **Tuning hyperparameters on test data** — Data leakage! Breaks your honest evaluation
3. **Trusting training accuracy** — Always check test performance
4. **Forgetting to scale** — Feature scaling matters here too
5. **No visualization** — Plot residuals, feature distributions, correlations

Prevention: Follow the pipeline. One step at a time. Visualize everything.

Today's Activity: Modeling Project

Work through **Activity 04: Modeling Project** — open-ended by design.

Step 1: Baseline (Expect Failure)

- Load the molecular energy dataset (1275 features)
- Build a simple Linear Regression with all features
- Check R^2 on test data. Probably bad (~0.2). That's fine — it's diagnostic.

Step 2: Explore & Diagnose

- Plot feature distributions: what's normal? What's skewed?
- Check feature correlations: are some redundant?
- Diagnose: where is your model failing?

Step 3: Try an Improvement